

Robust Predictions in Information-Acquisition Games

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MOTIVATION

Traditional information economics takes information as *given*.

However, agents often choose what information to acquire.

Two approaches for modeling information acquisition:

- Postulate a restricted set of available info structures.
- Make info acquisition flexible (aka, “rational inattention”).

Issue:

- Predictions depend on the assumed acquisition technology.
- Verifying assumptions on the acquisition technology is hard.

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We also apply our framework for “robust policy/design.”

Information Acquisition Games

BASE GAME

A **base game** is a tuple, $G = (I, \Theta, \pi, A, (u_i)_{i \in I})$, consisting of

- a finite set I of players,
- a finite payoff-relevant state Θ (“payoff state”),
- a prior for the payoff state, $\pi \in \Delta(\Theta)$,
- a finite set of action profiles, $A = \prod_{i \in I} A_i$,
- a utility function $u_i : A \times \Theta \rightarrow \mathbb{R}$ for each player i .

Throughout, view G as known both to the players and the analyst.

INFORMATION TECHNOLOGIES

An **information technology** $T = ((Z, \zeta), (X_i, \mathcal{E}_i, C_i)_{i \in I})$ consists of

- a finite payoff-irrelevant state Z (“correlation state”),
- a conditional distribution for $z \in Z$, $\zeta : \Theta \rightarrow \Delta(Z)$,
- a finite set of signal realizations X_i for every i ,
- a set of feasible experiments for player i , $\mathcal{E}_i \subseteq (\Delta(X_i))^{Z \times \Theta}$,
- a cost function $C_i : \mathcal{E}_i \rightarrow \mathbb{R}_+$ such that $\inf C_i(\mathcal{E}_i) = 0$.

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INFORMATION ACQUISITION GAME

Together, (G, T) induce an **information acquisition game**:

1. Simultaneously, each i chooses an experiment, $\xi_i \in \mathcal{E}_i$.
2. Nature determines (z, θ) , draws signals, $(x_i)_{i \in I}$.
3. Each player i observes their signal x_i and takes an action a_i .
4. Each player i gets a payoff $u_i(a, \theta) - C_i(\xi_i)$.

Solution Concept: Nash.

Remark: Information acquisition is covert.

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This is the equilibrium **outcome**.

- A vector $v = (v_i)_{i \in I}$ consisting of each player's payoff,

$$v_i = \sum_{a, \theta} u_i(a, \theta) p(a, \theta) - C_i(\xi_i^*).$$

We refer to v as the equilibrium **value**.

METHODOLOGY

We fix a base game G and a **class** $\mathcal{T}$ of technologies T .

Characterize the equilibrium outcome-value pairs that can arise in (G, T) across all T in the specified class.

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Characterize the equilibrium outcome-value pairs that can arise in (G, T) across all T in the specified class.

We consider three classes of information technologies:

- All **exogenous** technologies (Bergemann and Morris 2016).
- All **costly** information technologies.
- All **flexible** technologies.

EXOGENOUS, COSTLY, AND FLEXIBLE TECHNOLOGIES

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- $\mathcal{E}_i$ contains an uninformative experiment.
- For $\xi_i, \xi'_i \in \mathcal{E}_i$, if $\xi_i \succ \xi'_i$, then $C_i(\xi_i) > C_i(\xi'_i)$.

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A technology T is **flexible** if it is costly, and if for all i ,

$$\mathcal{E}_i = (\Delta(X_i))^{\mathbb{Z} \times \Theta}$$

Results in an Example

INVESTOR-SPECULATOR EXAMPLE

$$\Theta = \{\theta_b, \theta_{avg}\}, \quad \pi(\theta_b) = \pi(\theta_{avg}), \quad I = \{1, 2\}, \quad A_1 = \{I, N\}, A_2 = \{S, N\},$$

and utilities are given by:

	S	N
I	-1, 1	-1, 0
N	0, -2	0, 0

θ_b

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$$p(I, a_2, \theta_b) = 0 \forall a_2, \quad 3p(I, S, \theta_{avg}) \geq 2p(S), \quad 3p(I, N, \theta_{avg}) \leq p(N).$$

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But the only p that arises with a **costly technology** is

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Moreover, we study **weak separation** as a refinement of **ex. info**:

- fixing G , **w-separation** refines **nothing** or **almost everything**.
- characterize when a refinement occurs.
- for a **generic** G , weak separation refines **nothing**.

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Can induce p with **exogenous info** iff $\sum_a p(a, \theta) = 1/2 \forall \theta$ and

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Moreover, if we perturb the game,

	S	N
I	$-1, 1$	$-1, 0$
N	$0, -2$	$0, 0$

θ_b

	S	N
I	$-\varepsilon, 1$	$-\varepsilon, 1$
N	$0, -2$	$0, 0$

θ_{avg}

then

	S	N
I	0	0
N	0	$1/2$

θ_b

	S	N
I	0	0
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θ_{avg}

is the only p that arises with **exogenous info** and so with **costly** info.

VALUES A MODIFIED INVESTOR-SPECULATOR EXAMPLE

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Now, the following p is consistent with **ex. info** and **costly info**:

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With **ex. info**: $v_S = 1/2$.

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With **ex. info**: $v_S = 1/2$. With **costly info**, can have any $v_i \in [0, 1/2)$.

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Theorem 2. Suppose p arises for some **costly** T . Then (p, v) arises for some **costly** T' iff for every i ,

$$\text{either } \underline{v}_i(p) = \bar{v}_i(p) = v_i, \quad \text{or } v_i \in [\underline{v}_i(p), \bar{v}_i(p)].$$

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Applications:

- Optimal policy in a regime change game.
- Robust informational simplicity.
- Mechanism design with aftermarket concerns.